Chromogenic medium is used for the simple and fast detection of transformed bacteria by using chromogenic substrates. The chromogenic mixture contains substrates such as Salmon-GAL, X-GAL. Certain bacterial enzymes cleave the chromogenic substrate resulting in the coloured colonies.

HOTS

Why is EMB medium called a selective, differential as well as complex medium?

v) Enrichment medium

Enrichment medium is a liquid medium. It is used to grow a particular microorganism that is present in much smaller number along with others present in sufficiently large numbers. An enrichment medium provides nutrients and environmental conditions that favour the growth of a desired microorganisms. It is used to culture microorganisms present in soil or faecal samples that are very small in number. Example: Selenite F Broth is used to isolate *Salmonella typhi* present in low density in faecal sample. It is cultured in an enrichment medium containing Selenium. Selenium supports the growth of the desired organism and increase it to detectable levels compared to intestinal flora. Sodium selenite inhibits many species of Gram positive and Gram negative bacteria including *Enterococci* and coliforms.

vi) Antibiotic sensitivity medium

Antibiotic sensitivity medium is a microbiological growth medium that is commonly used for antibiotic sensitivity testing. Example: Muller-Hinton agar medium. It is a non-selective and non-differential medium. It allows the growth of most type of microorganisms. It contains starch which absorbs toxins released from bacteria. Hence toxins do not interfere with antibiotics. Agar concentration of 1.7% is used in this media which allows better diffusion of antibiotics (Figure 5.9).

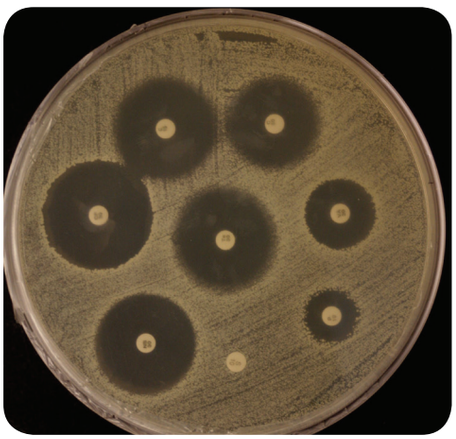


Figure 5.9: Antibiotic sensitivity on Muller Hinton agar

vii) Anaerobic medium

Anaerobic medium is a medium used for the cultivation of anaerobes, Example: i) Robertson cooked meat medium: This is used for the isolation of *Clostridium* ii) Thioglycolate broth: In this medium Sodium thioglycollate is used as a reducing agent which maintain a low Oxygen tension by removing the molecular Oxygen from the environment.

viii) Transport medium

Transport medium is used for the temporary storage of specimens that are being transported to the laboratory for cultivation. It maintains the viability of all

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